

Prajakta Dalawai

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SUMMARY

Aspiring AI Engineer with hands-on experience in Machine Learning, Generative AI, and IoT, complimented by an IEEE-accepted research publication at IRAI 2026, Melbourne, Australia. Committed to building scalable, data-driven AI solutions that solve real-world problems, while bringing proven leadership as General Secretary of the college student body.

INTERNSHIP EXPERIENCE

Machine Learning Intern, Fox Trading

Sep 2025 – Nov 2025

- Developed a Bike Sharing Demand Forecasting model using SVM and Linear Regression with feature selection and R^2 evaluation for real-world demand prediction.

Hybrid

EDUCATION

S G Balekundri Institute of Technology, B.E. in Computer Science & Engineering

2023 – 2027

CGPA: 9.01 / 10

Belagavi, Karnataka

SKILLS

Programming: Python, C++, Java

Web & Backend: HTML, CSS, FastAPI, REST APIs, MySQL, PostgreSQL(Have worked on Pgvector, basic)

ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, LangChain, LangGraph, RAG

Tools & Platforms: Git, GitHub, Power BI, AWS(basic), Linux (basic command line)

IoT & Embedded: Raspberry Pi, PLC

Soft Skills: Leadership, Communication, Teamwork

PROJECTS

Small Language Model | GitHub , Raspberry Pi 4, Phi-3 Mini, llama.cpp, ChromaDB, RAG, Python

- Developed an offline edge-native industrial SLM diagnostic assistant for air-gapped facilities.
- Benchmarked at RAG MRR 0.83, TTFT 2.4 s, peak RAM 3.41 GB, accuracy 93.1%, sustained 67.8 °C first domain-specific industrial benchmark on Raspberry Pi hardware.

NyayaGyan: AI-Powered Legal Assistant | GitHub , LLMs, LangChain (RAG), Neo4j, PostgreSQL pgvector, OCR, spaCy

- Built a role-based AI legal assistant using RAG and Knowledge Graphs to deliver accurate, citable insights.
- Implemented a multi-user web platform for document drafting, case-law retrieval, and hybrid search combining BM25 and embeddings.

GMNE: Generative Multi-Physics Neural Emulator for Hydrogen Electrolyzers | GitHub ,

PyTorch, FastAPI, NumPy, Pandas, Scikit-learn, Google Gemini API

- Addressed the high latency and cost of CFD/FEM-based PEM electrolyzer simulations (hours per run), which limit real-time control and digital twin deployment in hydrogen systems.
- Developed a PINN(Physics Informed Neural networks) and DDPM model, achieving $R^2 > 0.94$ with $<2\%$ error and provided uncertainty estimates (95% CI) with inference latency <100 ms

RESEARCH PUBLICATIONS

- *Adaptive On-Device Small Language Models for Offline Industrial Maintenance Assistance with Generative Fault Scenario Synthesis* — accepted at IEEE IRAI 2026, Melbourne, Australia.
- *AI-Powered Legal Assistant "NyayaGyan"* — published with DOI.
- Co-authored two research manuscripts for the IEEE IES Generative AI Challenge (IRAI/IECON).

ACHIEVMENTS AND AWARDS:

- **National-Level Hackathon Winner — IEEE NeoNexus 36.0**
- Cleared **Google's The Big Code Qualifier** — Top 15,000 Globally
- Secured 99.5 Percentile (AIR 2346) — **AINCAT 2026** among 4.7 lakh+ participants
- Accepted **Contributor — GSSoC 2026** (AI/Agents & Open Source Track)
- **Balayogini Award** recipient
- Selected as researcher at **STEAM-H Training and Research Foundation** (Robotics, AI, Additive Manufacturing, and Industry 4.0)

EXTRACURRICULARS

- General Secretary, College Student Body
- National-Level Karate Gold Medalist
- NSS State-Level Conference — 1st Place
- National Level Talk Masters — 1st Place
- Public Speaking, Debate & Elocution competitions